PeerLearn: Unit Test, Code Coverage and Branch Coverage Report

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Result: all 61 automated tests passed (0 failed) in 8 test suites. Line coverage is 94.9% and branch coverage is 80.4%, above the 70% line-coverage target in NFR-08.

1. Test run

Item	Value
Date of run	31 Oct 2026, 09:58 IST
Code version	`main` branch, commit `cd6df6f` (github.com/amoghshetty05/peerlearn)
Test framework	Jest 29 with Supertest (HTTP API tests)
Environment	Node.js 22, PostgreSQL 16 test database created fresh for each run
Command	`cd backend && npm test` (runs `jest --runInBand --coverage`)
Automation	The same tests run in GitHub Actions (job `backend`) on every pull request and every push to `main`; the job fails if line coverage drops below 70%

2. Summary

Measure	Result
Test suites	8 passed of 8
Test cases	61 passed, 0 failed, 0 skipped
Total test time	16.5 s
Statement coverage	92.0% (837/910)
Branch coverage	80.4% (288/358)
Function coverage	86.9% (205/236)
Line coverage	94.9% (723/762) (target 70%, NFR-08)

3. Unit test report

Two suites test single functions in isolation (`tests/unit`). The other six test each API endpoint against a real test database (`tests/integration`). Test names start with the Test Plan case they verify (TC-01 to TC-19).

Suite	Type	Tests	Passed	Failed	Time (s)
`tests/unit/scoring.test.js`	Unit	6	6	0	0.03
`tests/unit/time.test.js`	Unit	4	4	0	0.04
`tests/integration/auth.test.js`	Integration	18	18	0	5.78
`tests/integration/groups-moderation.test.js`	Integration	7	7	0	1.18
`tests/integration/performance.test.js`	Integration	1	1	0	2.58
`tests/integration/profile-matching.test.js`	Integration	8	8	0	1.75
`tests/integration/sessions.test.js`	Integration	11	11	0	2.89
`tests/integration/validation.test.js`	Integration	6	6	0	2.28

3.1 `tests/unit/scoring.test.js`

#	Group	Test	Result	Time (ms)
1	match score (FR-06, FR-15)	full marks without rating bonus is 100	Passed	1
2	match score (FR-06, FR-15)	availability: covers 30, same day 15, none 0	Passed	0
3	match score (FR-06, FR-15)	proficiency maps to 7 / 13 / 20 of the 20 points	Passed	1
4	match score (FR-06, FR-15)	rating bonus needs 3 ratings and equals avg/5*10	Passed	0
5	match score (FR-06, FR-15)	total is capped at 100	Passed	0
6	match score (FR-06, FR-15)	rank returns at most 5, descending, ties broken by rating count then id	Passed	1

3.2 `tests/unit/time.test.js`

#	Group	Test	Result	Time (ms)
7	-	localParts converts to IST ISO weekday and minutes	Passed	1

#	Group	Test	Result	Time (ms)
8	-	toIst formats with +05:30 like the SAD example	Passed	0
9	-	nextOccurrences returns the next two weekly starts after a moment	Passed	1
10	-	hhmmToMinutes	Passed	0

3.3 `tests/integration/auth.test.js`

#	Group	Test	Result	Time (ms)
11	TC-01 / TC-02 registration (FR-01)	TC-01 valid registration returns 201, no password, and the user can log in	Passed	645
12	TC-01 / TC-02 registration (FR-01)	tutor role can be chosen; admin cannot self-register	Passed	328
13	TC-01 / TC-02 registration (FR-01)	TC-02 non-university email is rejected and no account is created	Passed	36
14	TC-01 / TC-02 registration (FR-01)	short password (boundary 7 vs 8) and duplicate email	Passed	321
15	TC-01 / TC-02 registration (FR-01)	uppercase email is treated as the same address	Passed	636
16	TC-03 login (FR-02, SR-04)	correct password gives 200 and a token; wrong gives 401 and no token	Passed	468
17	TC-03 login (FR-02, SR-04)	logout ends the session: the token stops working	Passed	166
18	TC-04 role restriction (SR-05)	Learner and Tutor tokens get 403 on admin endpoints; report is unchanged	Passed	84
19	TC-04 role restriction (SR-05)	learner-only and tutor-only endpoints reject the other role	Passed	48
20	TC-19 token and HTTPS (SR-01, SR-04)	no token and wrong token get 401 UNAUTHENTICATED	Passed	41
21	TC-19 token and HTTPS (SR-01, SR-04)	a 3-hour-old token gets 401 TOKEN_EXPIRED	Passed	43
22	TC-19 token and HTTPS (SR-01, SR-04)	tokens live for 2 hours	Passed	103
23	TC-19 token and HTTPS (SR-01, SR-04)	plain HTTP is redirected to HTTPS when enforcement is on	Passed	37
24	TC-14 lockout and rate limit (SR-08, NFR-04)	5 wrong passwords lock the account; the correct password then gets 423	Passed	408
2	TC-14 lockout and rate	lock expires after 15 minutes and a	Passed	504

#	Group	Test	Result	Time (ms)
5	limit (SR-08, NFR-04)	successful login resets the counter		
26	TC-14 lockout and rate limit (SR-08, NFR-04)	the 101st request in a minute gets 429; other users are unaffected	Passed	323
27	TC-12 password storage and no secrets in responses (NFR-01, SR-02)	hash is bcrypt cost >= 10 and no endpoint returns hash or token fields	Passed	660
28	anonymous rate limit	anonymous traffic is limited per IP at a higher ceiling than the per-user limit	Passed	463

3.4 `tests/integration/groups-moderation.test.js`

#	Group	Test	Result	Time (ms)
29	TC-10 group capacity (FR-10)	first 3 join, 4th gets GROUP_FULL, after a leave the 4th can join	Passed	124
30	TC-10 group capacity (FR-10)	size boundaries 3..10 and the last seat is not double-booked under concurrency	Passed	98
31	TC-10 group capacity (FR-10)	admins cannot use group endpoints; list filters by subject	Passed	45
32	TC-11 report and suspend (FR-12)	report stored, admin suspends, tutor cannot log in, admin action logged	Passed	126
33	TC-11 report and suspend (FR-12)	warn notifies the user and counts a warning; dismiss changes nothing else	Passed	77
34	TC-11 report and suspend (FR-12)	cannot report yourself or a missing user; reason length enforced	Passed	40
35	TC-16 audit log (NFR-06, SR-07)	login, booking and admin action are recorded; edit and delete are refused	Passed	201

3.5 `tests/integration/performance.test.js`

#	Group	Test	Result	Time (ms)
36	-	TC-18 (short form): 100 concurrent users on GET /dashboard and POST /requests, 95% under 1 second	Passed	2136

3.6 `tests/integration/profile-matching.test.js`

#	Group	Test	Result	Time (ms)
37	TC-05 subjects and availability (FR-03, FR-04)	overlap gives 409 SLOT_OVERLAP; touching slots are fine; 11th subject rejected	Passed	116

#	Group	Test	Result	Time (ms)
38	TC-05 subjects and availability (FR-03, FR-04)	slot length boundaries: 30 and 120 minutes pass, 29 and 121 fail	Passed	56
39	TC-05 subjects and availability (FR-03, FR-04)	edit and delete a subject; subjects are normalised; invalid proficiency rejected	Passed	60
40	TC-06 ranked matches (FR-05, FR-06, FR-07, FR-15)	exactly 5 tutors, descending, none without maths, capped at 100, bonus only for the rated tutor	Passed	68
41	TC-06 ranked matches (FR-05, FR-06, FR-07, FR-15)	availability: slot covering the time scores higher than same weekday, higher than other day	Passed	44
42	TC-06 ranked matches (FR-05, FR-06, FR-07, FR-15)	suspended tutors are excluded; other learners cannot read the matches	Passed	54
43	TC-06 ranked matches (FR-05, FR-06, FR-07, FR-15)	help request validation (FR-05)	Passed	56
44	TC-13 matching speed (NFR-03)	20 match requests with 1,000 users and 200 tutors each return in under 2 seconds	Passed	813

3.7 `tests/integration/sessions.test.js`

#	Group	Test	Result	Time (ms)
45	TC-07 booking and contact privacy (FR-08, FR-11, FR-14, SR-03)	contact hidden until acceptance, then visible; both users notified	Passed	253
46	TC-07 booking and contact privacy (FR-08, FR-11, FR-14, SR-03)	decline notifies the learner and reveals nothing	Passed	87
47	TC-07 booking and contact privacy (FR-08, FR-11, FR-14, SR-03)	booking rules: outside availability, past, duplicate, wrong subject, not your request	Passed	90
48	TC-07 booking and contact privacy (FR-08, FR-11, FR-14, SR-03)	only the addressed tutor can answer	Passed	74
49	TC-07 booking and contact privacy (FR-08, FR-11, FR-14, SR-03)	a tutor cannot accept two sessions at the same time	Passed	87
50	TC-07 booking and contact privacy (FR-08, FR-11, FR-14, SR-03)	a failing email service never cancels the booking and is retried later	Passed	62

#	Group	Test	Result	Time (ms)
51	TC-08 request expiry (FR-08, FR-11)	after 49 hours the expiry job marks it expired and notifies the learner	Passed	74
52	reminders and completion (FR-11)	one hour before start both users get a reminder, once	Passed	89
53	reminders and completion (FR-11)	accepted sessions become completed after their start time	Passed	86
54	reminders and completion (FR-11)	notifications can be marked read	Passed	75
55	TC-09 ratings (FR-09) and tutor dashboard rating (FR-13)	pending rejected, 5 accepted, repeat 409, 6 gives 400	Passed	97

3.8 `tests/integration/validation.test.js`

#	Group	Test	Result	Time (ms)
56	TC-15 injection and validation (SR-06)	SQL injection in email and topic fields: no leak, tables unchanged	Passed	1297
57	TC-15 injection and validation (SR-06)	10,000-character topic is rejected with VALIDATION_ERROR	Passed	35
58	TC-15 injection and validation (SR-06)	wrong types, huge bodies, malformed JSON and bad ids all give the standard error shape	Passed	56
59	TC-15 injection and validation (SR-06)	internal errors return INTERNAL_ERROR with a request id and no details	Passed	41
60	TC-15 injection and validation (SR-06)	registration checks name length and phone format; profile update works	Passed	336
61	security headers	helmet headers present and X-Powered-By hidden	Passed	35

4. Code coverage report

Coverage shows how much of the backend code (`backend/src`, excluding the start-up file `server.js`) the tests executed. The full interactive report is in `docs/reports/coverage/index.html`.

File	Statements	Functions	Lines
`src/app.js`	100.0%	100.0%	100.0%
`src/config.js`	80.0%	100.0%	87.5%
`src/dal/audit.js`	80.0%	50.0%	80.0%

File	Statements	Functions	Lines
src/dal/groups.js	100.0%	100.0%	100.0%
src/dal/notifications.js	94.7%	88.9%	100.0%
src/dal/pool.js	100.0%	66.7%	100.0%
src/dal/reports.js	100.0%	100.0%	100.0%
src/dal/requests.js	85.7%	66.7%	85.7%
src/dal/sessions.js	81.4%	87.5%	85.4%
src/dal/tutors.js	96.5%	92.8%	96.4%
src/dal/users.js	95.8%	90.9%	100.0%
src/errors.js	100.0%	100.0%	100.0%
src/jobs/scheduler.js	0.0%	0.0%	0.0%
src/middleware/authenticate.js	100.0%	100.0%	100.0%
src/middleware/authorize.js	100.0%	100.0%	100.0%
src/middleware/enforceHttps.js	100.0%	100.0%	100.0%
src/middleware/errorHandler.js	89.5%	100.0%	88.2%
src/middleware/rateLimiter.js	87.1%	75.0%	92.3%
src/middleware/requestId.js	100.0%	100.0%	100.0%
src/middleware/validate.js	96.2%	100.0%	95.2%
src/routes/index.js	92.3%	85.3%	96.9%
src/schemas.js	100.0%	100.0%	100.0%
src/serializers.js	100.0%	100.0%	100.0%
src/services/audit.js	100.0%	100.0%	100.0%
src/services/auth.js	93.2%	100.0%	94.7%
src/services/dashboard.js	100.0%	100.0%	100.0%
src/services/email.js	100.0%	100.0%	100.0%
src/services/group.js	93.1%	100.0%	100.0%
src/services/matching.js	96.4%	85.7%	100.0%
src/services/moderation.js	94.7%	100.0%	100.0%
src/services/notification.js	94.1%	75.0%	100.0%
src/services/profile.js	85.9%	84.6%	95.5%
src/services/scoring.js	100.0%	100.0%	100.0%
src/services/session.js	96.0%	85.7%	98.8%
src/utils/clock.js	100.0%	100.0%	100.0%
src/utils/logger.js	80.0%	75.0%	75.0%
src/utils/time.js	100.0%	100.0%	100.0%
Total	**92.0%**	**86.9%**	**94.9%**

5. Branch coverage report

Branch coverage counts each possible path through an `if`, `? :`, `||` or `switch` and checks whether the tests took it. Overall, 288 of 358 branches (80.4%) were exercised. Files are listed from lowest to highest branch coverage.

File	Branches covered	Branch coverage
`src/dal/audit.js`	1 of 7	14.3%
`src/app.js`	1 of 2	50.0%
`src/dal/pool.js`	1 of 2	50.0%
`src/dal/reports.js`	2 of 4	50.0%
`src/services/email.js`	3 of 6	50.0%
`src/services/notification.js`	1 of 2	50.0%
`src/dal/sessions.js`	6 of 10	60.0%
`src/errors.js`	3 of 5	60.0%
`src/utils/logger.js`	7 of 11	63.6%
`src/dal/groups.js`	4 of 6	66.7%
`src/utils/time.js`	6 of 9	66.7%
`src/dal/tutors.js`	3 of 4	75.0%
`src/middleware/requestId.js`	3 of 4	75.0%
`src/services/profile.js`	24 of 32	75.0%
`src/schemas.js`	7 of 9	77.8%
`src/middleware/validate.js`	24 of 30	80.0%
`src/serializers.js`	8 of 10	80.0%
`src/services/group.js`	8 of 10	80.0%
`src/dal/users.js`	5 of 6	83.3%
`src/middleware/enforceHttps.js`	5 of 6	83.3%
`src/middleware/rateLimiter.js`	12 of 14	85.7%
`src/services/auth.js`	12 of 14	85.7%
`src/middleware/errorHandler.js`	13 of 15	86.7%

File	Branches covered	Branch coverage
`src/services/moderation.js`	14 of 16	87.5%
`src/config.js`	25 of 28	89.3%
`src/services/session.js`	42 of 47	89.4%
`src/services/scoring.js`	11 of 12	91.7%
`src/dal/notifications.js`	3 of 3	100.0%
`src/dal/requests.js`	0 of 0	100.0%
`src/jobs/scheduler.js`	0 of 0	100.0%
`src/middleware/authenticate.js`	14 of 14	100.0%
`src/middleware/authorize.js`	2 of 2	100.0%
`src/routes/index.js`	4 of 4	100.0%
`src/services/audit.js`	0 of 0	100.0%
`src/services/dashboard.js`	4 of 4	100.0%
`src/services/matching.js`	10 of 10	100.0%
`src/utils/clock.js`	0 of 0	100.0%
Total	**288 of 358**	**80.4%**

Files below 70% branch coverage: 11. The untested branches are mainly optional variations the tests do not need: database queries with and without an optional filter (for example listing audit records for one user or for all users), default values, and settings for a real SMTP server, which the tests replace with a simulated one. Every functional requirement is still covered by at least one passing test case.

6. How to reproduce

1. Start PostgreSQL 16 locally (or use Docker).
2. `cd backend && npm ci`
3. `TEST_ADMIN_URL=postgres://postgres:postgres@localhost:5432/postgres npm test`

In GitHub, open the Actions tab, choose any run of the CI workflow and download the `coverage` artifact for the same HTML coverage report.